A POWER-TORSION MODULE EMBEDS INTO A SUM OF INJECTIVE HULLS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The identification of the injective hull of a direct sum of copies of the residue field. Used inside the counted proof (source lines 1032-1035). Gives the ambient injective module of the embedding.
- The characterization of essential extensions through non-annihilated multiples. Invoked inside the counted proof (source lines 1035-1037). Shows that the maximal-ideal torsion sits essentially in the module.

Source reference. Original lemma lemma-essential-extension: sources/stacks-dualizing/pinned-original-source.tex, source lines 371–371. Navigation pointer only: the counted text refers to the Stacks item lemma-essential-extension, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let $(R, \mathfrak{m}, \kappa)$ be a local ring. Let E be the injective hull of κ. Let M be a $\mathfrak{m}$-power torsion R-module with $n = \dim_\kappa(M[\mathfrak{m}]) < \infty$. Then M is isomorphic to a submodule of $E^{\oplus n}$.*

Proof. Observe that $E^{\oplus n}$ is the injective hull of $\kappa^{\oplus n} = M[\mathfrak{m}]$. Thus there is an R-module map $M \to E^{\oplus n}$ which is injective on $M[\mathfrak{m}]$. Since M is $\mathfrak{m}$-power torsion the inclusion $M[\mathfrak{m}] \subset M$ is an essential extension (for example by Lemma lemma-essential-extension) we conclude that the kernel of $M \to E^{\oplus n}$ is zero. □

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